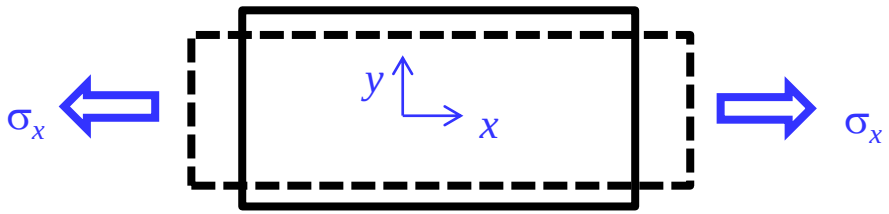
Chapter 9. Stress and strain in more than one direction

9.1. Introduction

In many situations, the structures/materials are subjected to complicate external loads. It is, therefore, important to obtain the important relationships involving a material's properties that are used when the material is subjected to multiaxial stress and strain. To do so we will assume that the material is ***homogeneous*** and ***isotropic*** and behaves in a linear-elastic manner.

9.2. Poisson's Ratio



υ is a material property with typical values are 0.25 to 0.4, never higher than 0.5.

For structural steel, υ ~ 0.32.

• For a body subjected to axial tension ***only***:

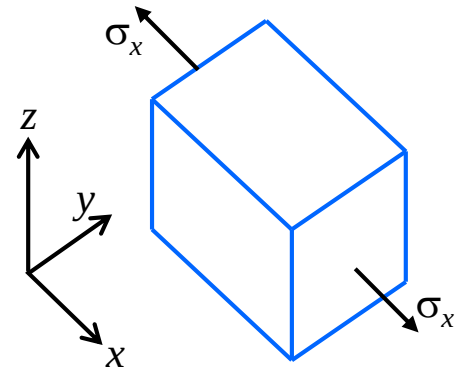
$$\sigma_y = 0,\ \sigma_z = 0\ ,\ \varepsilon_x = \frac{\sigma_x}{E} \qquad (9.1)$$

• The elongation in the axial direction is accompanied by a contraction in the radial directions:

$$\varepsilon_y = -\upsilon\varepsilon_x \qquad (9.2)$$

• Where υ (Greek letter, nu) is the Poisson's ratio

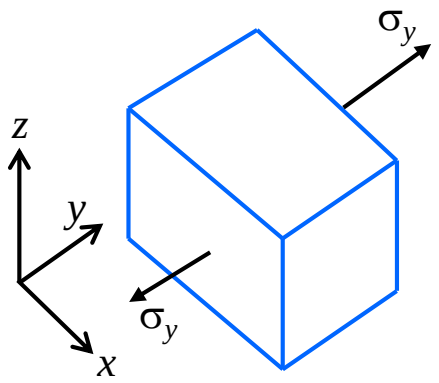
• Like Young's modulus *E* and shear modulus *G*, υ is also a mechanical property of material.



A body has a stress in the *x*-direction *only*.

The strain in the *x* direction will be: $\varepsilon_x = \frac{\sigma_x}{E}$

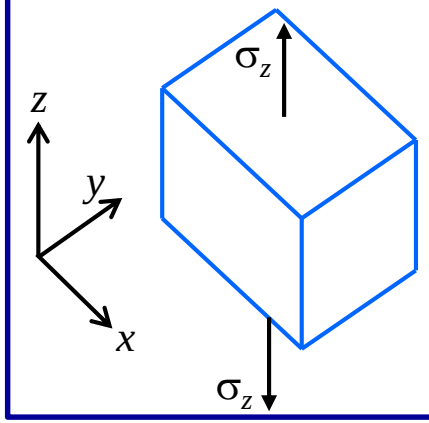
The strains in the *y* and *z* directions are: $\varepsilon_y = \varepsilon_z = -\upsilon\varepsilon_x = -\upsilon\frac{\sigma_x}{E}$



A stress in the *y*-direction *only*

$$\varepsilon_y = \frac{\sigma_y}{E}$$

$$\varepsilon_x = \varepsilon_z = -\upsilon\varepsilon_y = -\upsilon\frac{\sigma_y}{E}$$



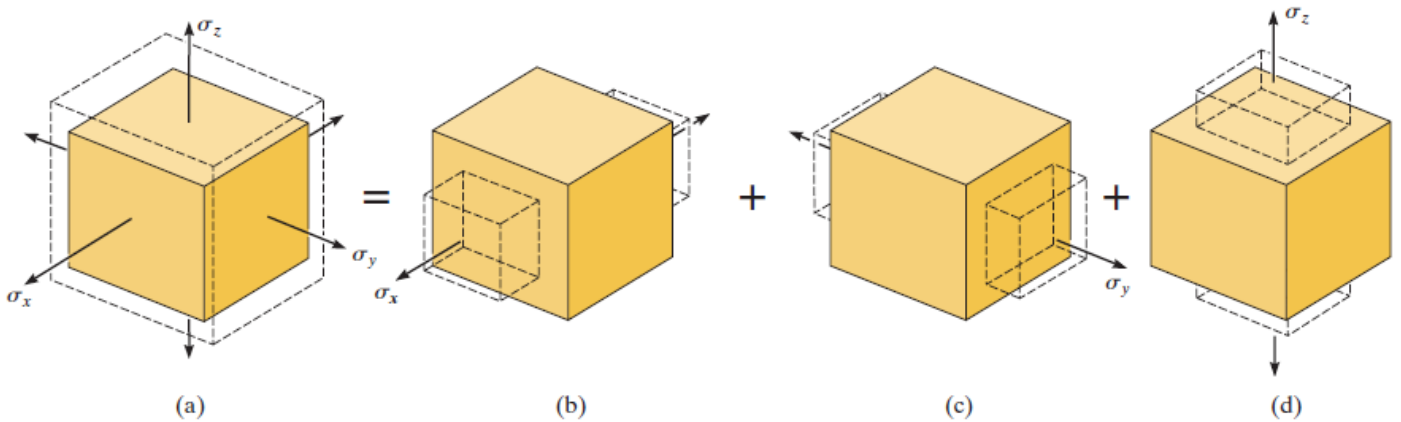
A stress in the *z*-direction *only*

$$\varepsilon_z = \frac{\sigma_z}{E}$$

$$\varepsilon_x = \varepsilon_y = -\upsilon\varepsilon_z = -\upsilon\frac{\sigma_z}{E}$$

9.3. Multi-axial loading

- Use the principle of superposition

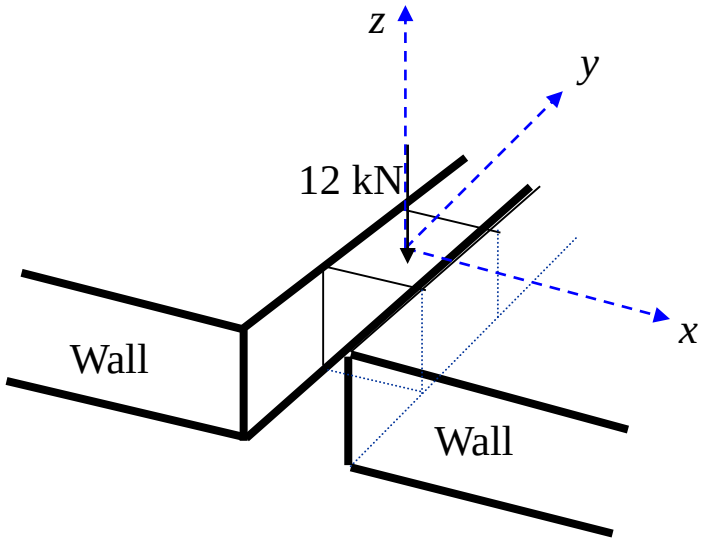


$$\varepsilon_x = \frac{\sigma_x}{E} - \upsilon\frac{\sigma_y}{E} - \upsilon\frac{\sigma_z}{E}$$

$$\varepsilon_y = -\upsilon\frac{\sigma_x}{E} + \frac{\sigma_y}{E} - \upsilon\frac{\sigma_z}{E} \qquad (9.3)$$

$$\varepsilon_z = -\upsilon\frac{\sigma_x}{E} - \upsilon\frac{\sigma_y}{E} + \frac{\sigma_z}{E}$$

Example 1. 12 kN forces are applied to the top and bottom of a cube (20 mm edges), which is sandwiched between two walls. Take E = 60 GPa and υ = 0.3. Find (i) the force exerted by the walls, (ii) ε_y.



Solution

This is a multi-axial loading problem and we use Eq. (6.3) to obtain the solution.

By observation, we see that $\varepsilon_x = 0$, $\sigma_y = 0$ and

$$\sigma_z = \frac{-12{,}000}{20 \times 20} = -30 \text{ MPa}$$

Putting these into the first of Eq. (6.3), one obtains

$$0 = \frac{\sigma_x}{60{,}000} - 0.3\frac{0}{60{,}000} - 0.3\frac{-30}{60{,}000}$$

$$\Rightarrow \sigma_x = -9 \text{ MPa}$$

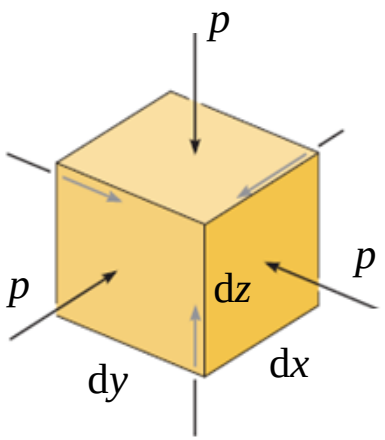
(i) The force exerted by the walls is

$$F = A\sigma_x = (20 \times 20)(-9) = -3{,}600 \text{ N}$$

(ii) ε_y is calculated from the second of Eq. (6.3) as

$$\varepsilon_y = -0.3\frac{-9}{60{,}000} + \frac{0}{60{,}000} - 0.3\frac{-30}{E} = 1.95 \times 10^{-4}$$

9.4. Hydrostatic loading



Hydrostatic stress

For special case of “hydrostatic” loading, the body is subjected to equal compressions in three directions. From Eq. (6.3), we get

$$\varepsilon_x = \varepsilon_y = \varepsilon_z = -p\frac{1-2\upsilon}{E}$$

For stable materials, υ can not be larger than 0.5. Otherwise, compressive stresses in all three directions will result in the expansion of the body. This is physically not possible.